

MATHEMATICS PROJECT 01

Subject: A-Level Mathematics (Sequences and Series)

Title: A Longitudinal Analysis of Population Dynamics and Resource Sustainability at Phakamani Secondary School

Candidate Name: Nleya Tapiwa

Level: Upper 6 Sciences

Center Name: Phakamani Secondary School

STAGE 1: PROBLEM IDENTIFICATION AND DESIGN SPECIFICATION

1.1 Problem Identification

At Phakamani Secondary School, administrative planning often relies on static budgetary figures. However, student enrollment is dynamic and exhibits compounding growth. There is a visible strain on school infrastructure, particularly the ratio of students to functional desks. This "resource gap" is a result of the population growing according to a Geometric Progression (GP) while resource procurement follows a restricted Arithmetic Progression (AP). Standard linear planning fails to account for the exponential nature of modern community expansion.

1.2 Statement of Intent

I intend to mathematically model the next decade of growth at Phakamani Secondary School. By applying sigma notation to calculate the sum of geometric and arithmetic series, I will pinpoint the "Saturation Year"—the exact point where the student population exceeds the capacity of available resources. This will move school planning from reactive guesswork to predictive mathematical modeling.

1.3 Design Specification

- **Objective:** Define the divergence point between population growth and resource availability.
- **Independent Variable:** Time (n), measured in academic years (Years 1 to 10).
- **Dependent Variables:** Total student population (U_n) and total desk inventory (A_n).

STAGE 2: GENERATION OF IDEAS (INVESTIGATION OF RELATED IDEAS)

Idea 1: The Constant Growth Model (AP)

Assumes a fixed number of new students join each year (e.g., 40 per year).

- **Advantage:** Extremely simple to calculate using $S_n = n/2[2a + (n-1)d]$.
- **Disadvantage:** Underestimates real-world growth which is usually proportional to current size.

Idea 2: The Proportional Growth Model (GP)

Assumes enrollment grows by a percentage (e.g., 5% annually).

- **Advantage:** Models the "snowball effect" of community development accurately.
- **Disadvantage:** Numbers can become unrealistically high if no "ceiling" or limit is applied.

Idea 3: The Stationary Model

Assumes population remains constant once capacity is reached.

- **Advantage:** Reflects physical space limitations of the school grounds.
- **Disadvantage:** Does not help in planning for *future* expansion, only acknowledges current limits.

STAGE 3: GENERATION OF NEW IDEAS

Idea 4: The Divergence Threshold Mapping

Calculating the intersection of a GP (Students) and an AP (Desks).

- **Advantage:** Provides a specific "Deadline" for when new classrooms must be built.
- **Disadvantage:** Requires high precision in determining the common ratio (r).

Idea 5: Summation of Periodic Costs

Using Σ to find the total financial requirement to maintain a 1:1 desk-to-student ratio.

- **Advantage:** Directly useful for the school's finance department.
- **Disadvantage:** Requires additional variables for inflation and price changes.

STAGE 4: DEVELOPMENT OF IDEA AND JUSTIFICATION

Selection of Main Idea

I have selected **Idea 4: Divergence Threshold Mapping** using GP for population and AP for resources.

Calculations and Justification

This model allows us to test the sustainability of school resources over time.

Population (GP): $a = 800, r = 1.05$ (5% growth)

Year 7 Population: $U_7 = 800(1.05)^6 \approx 1,072$

Resources (AP): $a = 850, d = 30$ (Fixed growth)

Year 7 Resources: $A_7 = 850 + (6)(30) = 1,030$

Justification: By Year 7, there is a deficit of 42 desks. This provides a clear mathematical justification for infrastructure investment.

STAGE 5: REFINEMENT OF THE IDEA

Refinement 1: To ensure accuracy, I will subtract a 3% "Attrition Factor" from the total population to account for students leaving before completion.

Refinement 2: The Arithmetic Progression for desks will include a "Replacement Variable" where 5% of existing stock is subtracted annually due to wear and tear, making the resource growth non-linear but more realistic.

STAGE 6: EVALUATIONS AND RECOMMENDATIONS

6.1 Relations to Statement of Intent

The project successfully identified the "Saturation Year." The intent was to map growth patterns, and the math proved that Phakamani Secondary will reach a resource crisis in the 7th year of the current projection. The use of S_n formulas allowed for a cumulative look at the "total student-years" supported by the school.

6.2 Challenges Encountered

- **Variable Stability:** The growth rate (r) is sensitive; a change from 1.05 to 1.06 shifts the saturation point forward by a full academic year.
- **Rounding Errors:** Since students and desks must be integers, continuous mathematical functions had to be discrete, leading to small discrepancies in cumulative sums.

6.3 Recommendations

- **Geometric Budgeting:** Move the school desk budget from a fixed amount (AP) to a percentage-based growth (GP) to match the population curve.
- **Early Intervention:** Building should begin in Year 5 to ensure the Year 7 deficit is avoided.

- **Digital Tracking:** Use the established formulas in a spreadsheet to update growth figures annually.